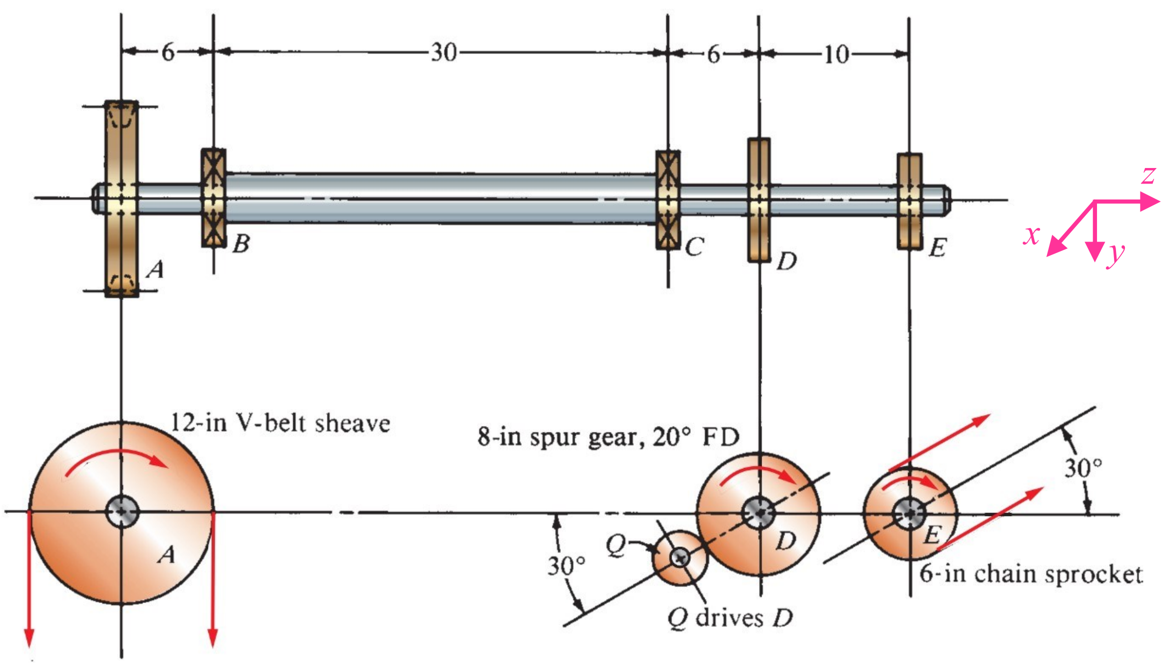
Machine Design (EMCH327 Section 001, Spring 2022)

Mid-Term Exam 3, *8:30 AM, Friday, Apr. 15th, 2022*

Name (First/Last): ______________________________ **Score:** ______________/100'

1. The shaft is part of a conveyor drive system that feeds crushed rock into a railroad car. The shaft rotates at 240 rpm and is subjected to moderate shock during operation. The gear at *D* receives 15 hp from pinion *Q*. The V-belt sheave at *A* delivers 10 hp vertically downward. The chain sprocket *E* delivers 5 hp. The shaft material is SAE 1137 cold-drawn steel. Take $C_s = 0.80$, $C_R = 0.81$, design factor $N = 4$, stress concentration factor $K_t = 3$. (**55'** in total)

1) Determine the torques on V-belt sheave *A*, gear *D*, and chain sprocket *E*. Also determine the torque distribution throughout the entire shaft.
2) Plot the shear and bending moment diagrams of the entire shaft in the horizontal and vertical planes.
3) Determine the total bending moment at point *C* and *D*.
4) Determine the minimum acceptable diameters for the shaft at **point *C***.



Solutions:

Step 1: Analyze the torque distribution in the shaft.

The total 15 hp power is split from *D* to the V-belt sheave *A* by 10 hp and then to the chain sprocket at *E* by the rest 5 hp.

Torque on gear *D*:

$$T_D = \frac{63000(15\text{ hp})}{240\text{ rpm}} = \boxed{3937.5\text{ lb}\cdot\text{in}}$$

Torque on V-belt sheave *A*:

$$T_A = \frac{63000(10\text{ hp})}{240\text{ rpm}} = \boxed{2625\text{ lb}\cdot\text{in}}$$

Torque on chain sprocket *E*:

$$T_E = \frac{63000(5\text{ hp})}{240\text{ rpm}} = \boxed{1312.5\text{ lb}\cdot\text{in}}$$

Using Free-Body-Diagram, we get the torque distribution throughout the entire shaft (note: both *B* and *C* are support bearings):

$$T_{A-B} = T_{B-C} = T_{C-D} = T_A = \boxed{2625\text{ lb}\cdot\text{in}}$$

$$T_{D-E} = T_E = \boxed{1312.5\text{ lb}\cdot\text{in}}$$

Step 2: Compute forces that act in the vertical and horizontal planes.

(1) Forces on V-belt sheave *A*:

Net driving force: $F_N = F_1 - F_2 = \frac{T_A}{\frac{D_A}{2}} = \frac{2625}{\frac{12}{2}} = 437.5\text{ lb}$

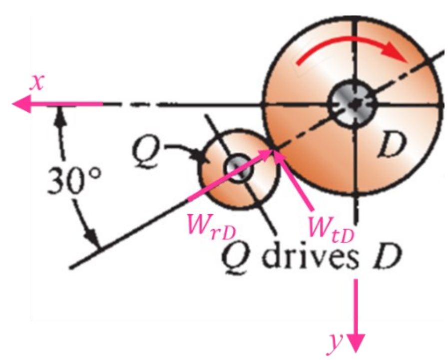
Bending force at *A*: $F_A = F_1 + F_2 = 1.5(F_1 - F_2) = 1.5(437.5) = 656.25\text{ lb}$ (downward)

The components of the bending force at point *A* are:

$F_{Ax} = 0\text{ lb}$

$F_{Ay} = 656.25\text{ lb}$ (downward or positive *y*-direction)

(2) Forces on gear *D*:



$$W_{tD} = \frac{T_D}{\frac{D_D}{2}} = \frac{3937.5}{8/2} = 984.4\text{ lb}$$

$$W_{rD} = W_{tD}\tan 20° = 984.4(\tan 20°) = 358.3\text{ lb}$$

The components of the bending force at point *C* are:

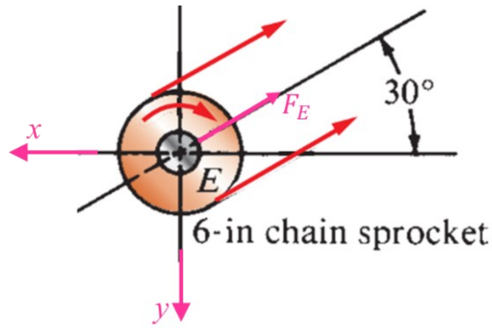
$$F_{Dx} = W_{tD}\sin 30° - W_{rD}\cos 30° = 984.4\sin 30° - 358.3\cos 30° = 181.9\text{ lb}$$

$F_{Dy} = -W_{tD}\cos 30° - W_{rD}\sin 30° = -984.4\cos 30° - 358.3\sin 30° = -1031.7 \text{ lb}$

(upward or negative y-direction)

(3) Forces on chain sprocket E:

$$F_E = \frac{T_E}{\frac{D_E}{2}} = \frac{1312.5}{6/2} = 437.5 \text{ lb}$$

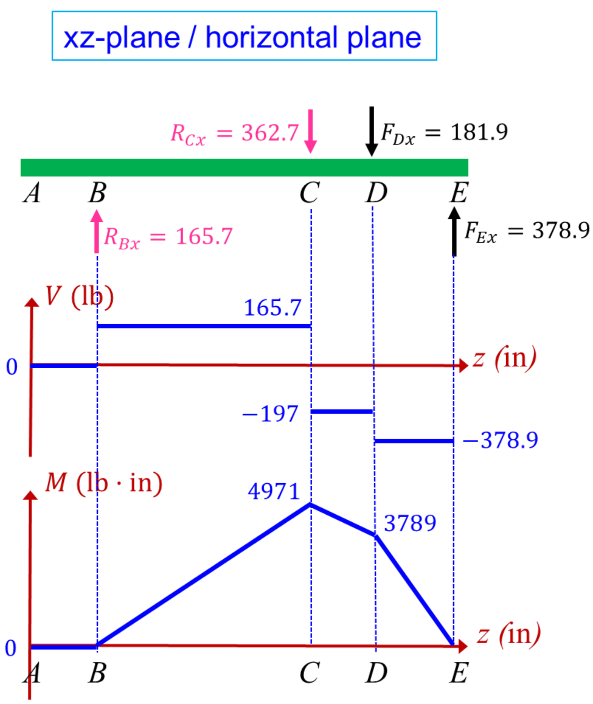


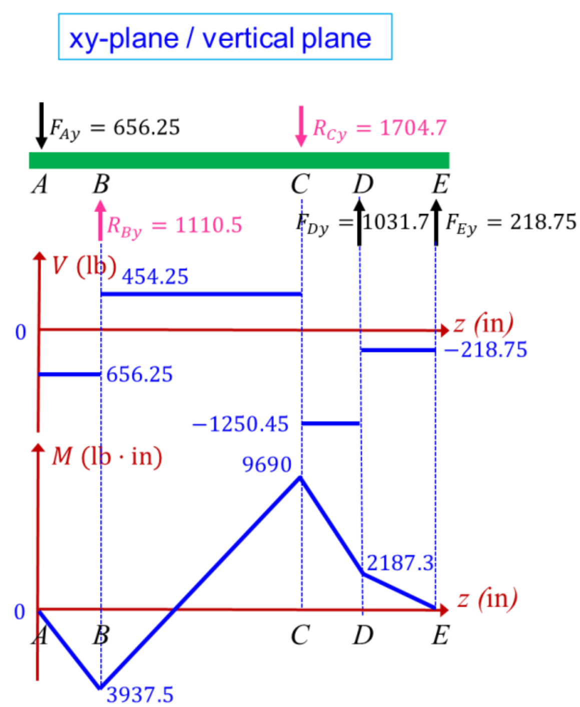
The components of the bending force at point E are:

$F_{Ex} = -F_{Ex}\cos 30° = -437.5\cos 30° = -378.9 \text{ lb}$
(negative x-direction)

$F_{Ey} = -F_{Ex}\sin 30° = -437.5\sin 30° = -218.75 \text{ lb}$
(upward or negative y-direction)

Step 3: Shear force and bending moment diagrams.





Total bending moment at point C:

$$M_C = \sqrt{4971^2 + 9690^2} = \boxed{10891 \text{ lb} \cdot \text{in}}$$

Total bending moment at point D:

$$M_D = \sqrt{3789^2 + 2187.3^2} = \boxed{4375 \text{ lb} \cdot \text{in}}$$

Step 4: Design shaft diameter at different places.

The material is SAE 1137 cold-drawn steel. From Appendix 3, we find

$s_y = 82$ ksi, $s_u = 98$ ksi

From Figure 5-11(a), we find the endurance limit $s_n = 37$ ksi. Then, the estimated actual endurance limit is

$s_n' = s_n C_m C_{st} C_R C_s = (37)(1.0)(1.0)(0.81)(0.80) = 23.976$ ksi $= 23976$ psi

For Point *C* on the **shaft**:

$T_C = T_{A-C} = T_{C-D} = 2625$ lb · in, $M_C = 10891$ lb · in

Using design equation for shaft under bending and torsion only, $D = [\frac{32N}{\pi}\sqrt{(\frac{K_t M}{s_n'})^2 + \frac{3}{4}(\frac{T}{s_y})^2}]^{1/3}$, with $K_t = 3$ and $N = 4$:

$$D_C = [\frac{32(4)}{\pi}\sqrt{(\frac{3(10891)}{23976})^2 + \frac{3}{4}(\frac{2625}{82000})^2}]^{1/3} = \boxed{3.82 \text{ in}}$$

2. A spring is composed of 9 total coils of a wire with a diameter of 0.15 in. The wire material is hard-drawn steel A227. The outer coil diameter is 1 in and the free length of the spring is 6 in. The ends of the spring are squared and ground. (**45'** in total)

1) Determine the maximum shear stress in the spring when a static load of 200 lb is applied to the spring.
2) Determine the deflection of the spring under this load of 200 lb.
3) Determine the spring rate.
4) Determine the solid length of the spring and the corresponding external load (assuming elastic deformation and there is no buckling) and maximum shear stress in the spring.
5) Will the spring buckle under the external load of 200 lb if the critical ratio is 0.26? How much the external load would be to let the spring buckle?

Solutions:

Given $D_w = 0.15$ in, $N = 9$, $OD = 1$ in, $L_f = 6$ in

1) In order to get the maximum shear stress using $\tau_{\max} = \frac{8KFD_m}{\pi D_w^3}$, we need to know the mean diameter of the spring (D_m), Wahl factor K which is a function of spring index C.

   Since outer coil diameter $OD = D_m + D_w$, we have:
   $D_m = OD - D_w = 1 - 0.15 = 0.85$ in

   Spring index $C = \frac{D_m}{D_w} = \frac{0.85}{0.15} = 5.667$
   Wahl factor $K = \frac{4C-1}{4C-4} + \frac{0.615}{C} = \frac{4(5.667)-1}{4(5.667)-4} + \frac{0.615}{5.667} = 1.269$
   Then, with external force $F = 200$ lb, we can get the maximum shear stress
   $$\tau_{\max} = \frac{8KFD_m}{\pi D_w^3} = \frac{8(1.269)(200)(0.85)}{\pi(0.15^3)} = \boxed{162771 \text{ psi}}$$

2) From Table 18-4, we find that, for wire material hard-drawn steel A227, the shear modulus $G = 11.5 \times 10^6$ psi.

   For squared and ground ends, the number of active coils is $N_a = N - 2 = 9 - 2 = 7$.

   Using the linear deflection equation $f = \frac{8FD_m^3N_a}{GD_w^4} = \frac{8FC^3N_a}{GD_w}$, we have:

   $$f = \frac{8FC^3N_a}{GD_w} = \frac{8(200)(5.667^3)(7)}{(11.5 \times 10^6)(0.15)} = \boxed{1.182 \text{ in}}$$

3) Spring rate $k = \frac{\Delta F}{\Delta L} = \frac{F}{f} = \frac{200}{1.182} = \boxed{169.2 \text{ lb/in}}$

4) The solid length is the total length when all coils are compressed in close contact, i.e.,

$$L_s = ND_w = 9(0.15) = \boxed{1.35 \text{ in}}$$

The corresponding external load is:

$$F = \frac{fGD_w}{8C^3N_a} = \frac{(6-1.35)(11.5\times 10^6)(0.15)}{8(5.667)^3(7)} = \boxed{787 \text{ lb}}$$

The corresponding maximum shear stress in the spring is:

$$\tau_{\max} = \frac{8KFD_m}{\pi D_w^3} = \frac{8(1.269)(787)(0.85)}{\pi(0.15^3)} = \boxed{640504 \text{ psi}}$$

5) Ratio of free length to mean diameter $\frac{L_f}{D_m} = \frac{6}{0.85} = 7.1 \approx 7.$

From Figure 18-15 Curve A, we find that, the critical ratio is about 0.26, i.e., $\frac{f_o}{L_f} = 0.26$, where f_o is the operating deflection and L_f is the free length.

Check our data: $f_o = 1.182$ in, $L_f = 6$ in, then

$$\frac{f_o}{L_f} = \frac{1.182}{6} = 0.197 < 0.26$$

So, the spring **will not buckle** under the external load of 200 lb.

To let the spring buckle, the deflection should reach to the critical ratio, i.e., $\frac{f_o{}'}{L_f} = 0.26$, then the new deflection is:

$$f_o' = 0.26L_f = 0.26(6) = 1.56 \text{ in}$$

Since we assume the spring deforms linearly to the external load, we have already known that, for external load of 200 lb, the deflection is 1.182 in, then using the Hooke's (linear) law, we get the new loading required for the deflection that lets the critical ratio reach 0.26

$$F' = \frac{1.56}{1.182}\times 200 = \boxed{264 \text{ lb}}$$

[*Bonus Problem***]** Devise a gear train using all external gears on parallel shafts. Use 20° full-depth involute teeth and no less than 18 teeth and no more than 160 teeth in any gear. The input speed is 6160 rpm and the output speed must be **exactly** 22 rpm. Ensure that there is no interference. Specify the number of teeth for each gear pair and sketch the layout for your design. (**10'** in total)

Solutions:

Given: $n_{\text{input}} = 6160$ rpm, $n_{\text{output}} = 22$ rpm, exact ratio required.

First, we compute the nominal train value (TV):

$TV = (\text{input speed})/(\text{output speed}) = 6160/22 = 280$

Factors are: $2 \times 2 \times 2 \times 5 \times 7$

For 20° full-depth involute teeth, $N_{\min} = 18,\ N_{\max} = 160$

Nominal maximum velocity ratio per pair:

$$VR_{\max} = \frac{N_{\max}}{N_{\min}} = \frac{160}{18} = 8.89$$

Single gear pair: $VR = 8.89$ (Too small)

Two pairs: $VR = (8.89)^2 = 79.03$ (Too small)

Three pairs: $VR = (8.89)^3 = 702.6$ (fulfill requirement, three pairs required)

Recombine VR factors: $VR_1 = 8, VR_2 = 7, VR_3 = 5$

Gear pair #1: $N_A = 18, N_B = 8 \times 18 = 144$

Gear pair #2: $N_C = 18, N_D = 7 \times 18 = 126$

Gear pair #3: $N_E = 18, N_F = 5 \times 18 = 90$

Check final $TV = \frac{144}{18} \times \frac{126}{18} \times \frac{90}{18} = 280$ **Exact, OK!**

The schematic gear train is shown on the right.